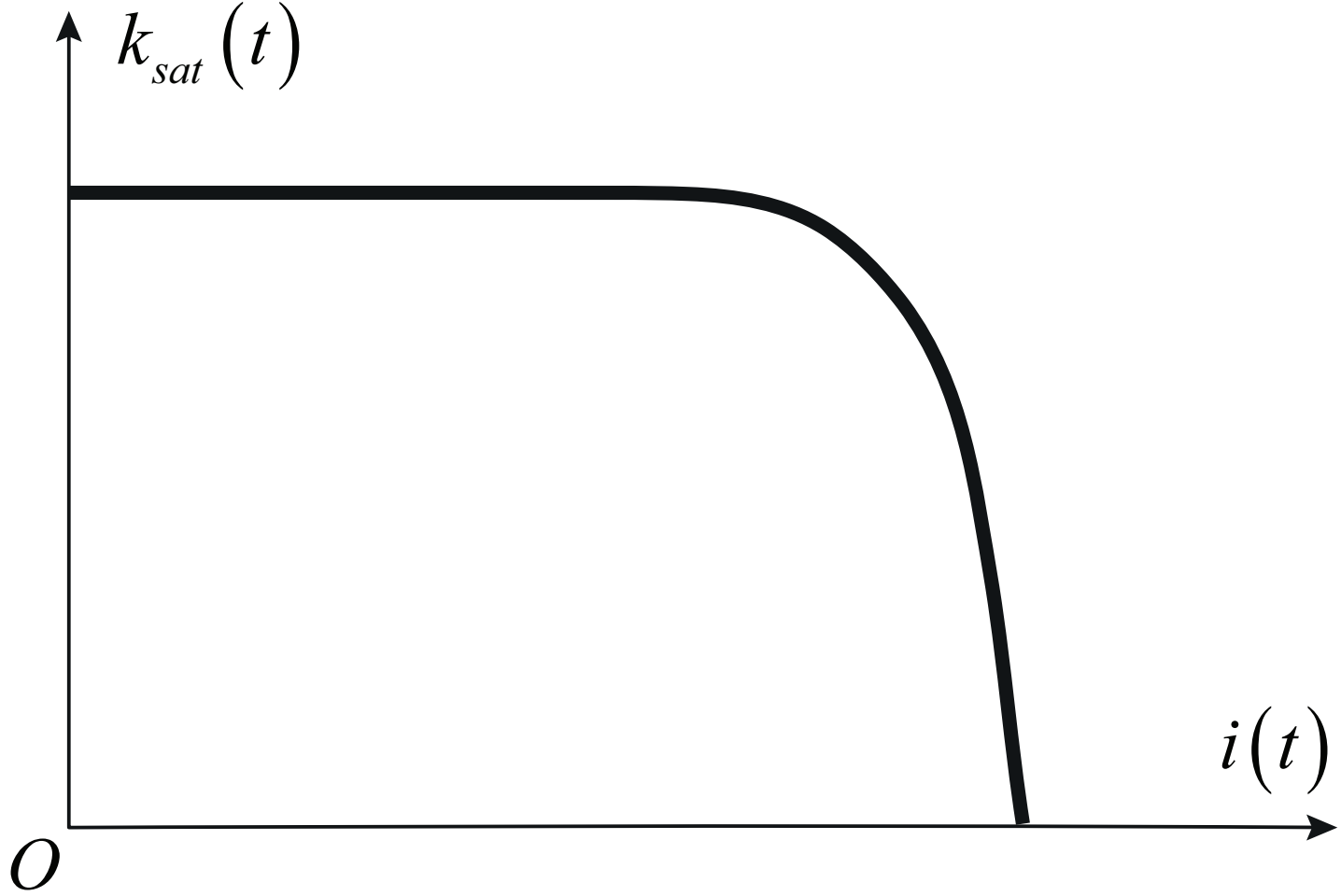
$k_{sat}(t)$
$i(t)$
$O$